

Emerging Radar Technology: Real-Time Through-the-Wall Surveillance



Sarath Mohan K.T.
is project manager at
Mistral Solutions Pvt Ltd

Imagine hostages being taken by terrorists in a large building with all modes of communication shut-off. Rescuers are ready to deploy, but cannot because they do not know where the hostages are inside the building. They now have to spend a considerable amount of precious time trying to locate the hostages.

But, what if they could see through the walls?

Through-the-wall surveillance (TWS) technology is an emerging technology that allows the user to see through obstacles, such as concrete walls, doors and other visually-opaque materials. Though initially developed for military use, it is

now being used by law-enforcement agencies as costs have become more affordable.

What is TWS

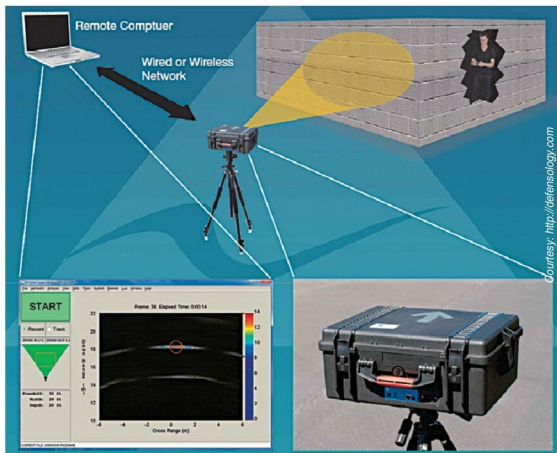
There are usually two types of devices that use this technology—portable hand-held devices that alert officers to the presence of an individual behind doors or walls, and portable, computer based devices that can help bigger security teams to visualise events during hostage situations.

Electromagnetic waves within the lower GHz range can penetrate most building materials, allowing for the detection of hidden targets. Every person or object can reflect an electromagnetic wave, which can be registered by the radar receiver. The challenge

is to distinguish between wanted (living persons) and unwanted objects (rubble or interiors of a room).

So, how does the device identify the presence of a human if there is no movement? A living being is always connected with some movements, like breathing, even when asleep or stationary. Breathing causes minor variations in the body's shape, which can be determined by the radar.

The radar is able to pinpoint and track people's movement behind a wall in real-time with down-range versus cross-range image of all targets. For recognition of static objects, radar imaging must be performed, which requires scanning and sophisticated data processing.



How TWS works